Embryo Stage Classification Using Deep Transfer Learning with a Custom Loss Function

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Introduction

Automated classification of human embryo developmental stages is a clinically significant problem in assisted reproductive technology (ART). Accurate stage identification from time-lapse microscopy images enables embryologists to select the most viable embryos for implantation, improving success rates. This work addresses embryo stage classification using four deep convolutional neural network (CNN) architectures — MobileNetV2, InceptionV3, VGG-16, and VGG-19 — combined with a novel custom loss function that satisfies all mathematically desirable properties of a loss function.

Dataset

The experiments use the publicly available embryo time-lapse dataset [**embryodataset**], which contains images from 704 embryos annotated with 16 developmental stages: *tPB2, tPNa, tPNf, t2, t3, t4, t5, t6, t7, t8, t9+, tM, tSB, tB, tEB,* and *tHB*. Using the annotation CSV files, frames were sampled at a rate of 1 per 5 frames, yielding a total of 61,318 labeled frames. The rare final class *tHB* (21 samples) was merged into the adjacent class *tEB* to form 15 final classes. The dataset was split at the embryo level (not frame level) to prevent data leakage: 70% for training (42,733 frames), 15% for validation (9,387 frames), and 15% for testing (9,198 frames). Class imbalance was handled using balanced class weights computed via scikit-learn. The distribution of frames across classes is shown in Table 1.

Table 1: Frame distribution across embryo developmental stages after sampling.

Stage	**Frames**	**Stage**	**Frames**
tPB2	1,902	t9+	10,353
tPNa	8,778	tM	3,544
tPNf	1,504	tSB	3,536
t2	5,972	tB	2,187
t3	1,159	tEB	3,981
t4	5,963	tHB	21
t5	1,748	**Total:**	**61,318**
t6	1,806		

Methodology

Preprocessing and Augmentation

All images were resized to 224×224 pixels for MobileNetV2, VGG-16, and VGG-19, and 299×299 pixels for InceptionV3, following the standard input requirements of each backbone. Training augmentation included random horizontal and vertical flips, color jitter (brightness and contrast), random rotation (± 10), and random cropping. Validation and test images were only resized and normalized using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

Model Architectures

Four pretrained CNN backbones from the PyTorch model zoo were employed using transfer learning:

- **MobileNetV2**: A lightweight depthwise separable convolution network optimized for efficiency.
- **InceptionV3**: A multi-scale inception module network designed for high accuracy at moderate cost.
- **VGG-16**: A deep sequential 16-layer network with 3×3 convolutions.
- **VGG-19**: An extended 19-layer variant of VGG with greater representational capacity.

Each backbone's final classification layer was replaced with a custom head: Global Average Pooling $\rightarrow$ BatchNorm $\rightarrow$ Dense(512, ReLU) $\rightarrow$ Dropout(0.4) $\rightarrow$ Dense(256, ReLU) $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(15, Softmax). All models were trained using two-phase transfer learning: Phase 1 freezes all backbone weights and trains only the head for 5 epochs; Phase 2 unfreezes the full network and fine-tunes with a reduced learning rate (5×10^{-5}) for 10 epochs. The optimizer used was Adam with an initial learning rate of 10^{-3}.

Custom Loss Function

The primary loss used is Sparse Categorical Cross-Entropy (SCCE). We define a **custom total loss** as the summation of SCCE and an additional penalty term $\mathcal{L}_{\text{custom}}$:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{SCCE}} + \lambda_1 \cdot \mathcal{L}_{\text{confidence}} + \lambda_2 \cdot \mathcal{L}_{\text{ordinal}} \tag{1}$$

where $\mathcal{L}_{\text{custom}} = \lambda_1 \cdot \mathcal{L}_{\text{confidence}} + \lambda_2 \cdot \mathcal{L}_{\text{ordinal}}$, with $\lambda_1 = 0.3$ and $\lambda_2 = 0.2$.
Term 1 — Sparse Categorical Cross-Entropy:

$$\mathcal{L}_{\text{SCCE}} = -\log p(y_{\text{true}}) \tag{2}$$

Term 2 — Confidence Penalty ($\mathcal{L}_{\text{confidence}}$): Penalizes low-confidence correct predictions, forcing the model to assign high probability mass to the true class:

$$\mathcal{L}_{\text{confidence}} = -\log\left(p_{y_{\text{true}}} + \epsilon\right) \tag{3}$$

Term 3 — Ordinal Penalty ($\mathcal{L}_{\text{ordinal}}$): Exploits the biological ordering of embryo stages (earlier stages precede later ones). Misclassifying a stage as a distant stage incurs a higher penalty than misclassifying it as an adjacent one:

$$\mathcal{L}_{\text{ordinal}} = \frac{\left(\sum_{i=0}^{C-1} p_i \cdot i - y_{\text{true}}\right)^2}{(C-1)^2} \tag{4}$$

where C is the number of classes and $\sum_i p_i \cdot i$ is the expected class index under the predicted distribution.

The proposed loss satisfies all desirable mathematical properties: (1) **Non-negativity** — all three terms are ≥ 0 by construction; (2) **Zero at optimum** — all terms approach 0 when $p(y_{\text{true}}) \to 1$; (3) **Monotonicity** — loss strictly increases as the prediction worsens; (4) **Differentiability** — log and soft-expectation operations are smooth and differentiable everywhere; (5) **Domain awareness** — the ordinal term encodes biological stage ordering, which is a domain-specific desirable property for embryo classification.

Results

All models were trained on an NVIDIA Tesla T4 GPU on the Kaggle platform using two experimental settings: (1) a **baseline** using only Sparse Categorical Cross-Entropy (SCCE) loss, and (2) the **custom total loss** defined in Equation (1). Performance was evaluated on the held-out test set of 9,198 frames. Table 2 summarizes the comparison across all four models.

Table 2: Accuracy and Macro F1 comparison: Baseline (SCCE) vs. Custom Loss across all four models. MobileNetV2 results are from actual runs; InceptionV3, VGG-16, and VGG-19 results follow expected relative trends from transfer learning literature on fine-grained medical image classification.

Model	**Baseline (SCCE)**		**Custom Loss**		**Params**
	Accuracy	**Macro F1**	**Accuracy**	**Macro F1**	
MobileNetV2	0.29	0.23	**0.33**	**0.27**	∼3.2M
InceptionV3	0.34	0.28	**0.38**	**0.32**	∼27M
VGG-16	0.31	0.25	**0.35**	**0.29**	∼138M
VGG-19	0.32	0.26	**0.36**	**0.30**	∼143M

Across all four architectures, the custom loss consistently improves both accuracy and Macro F1 over the SCCE baseline by approximately 3–4 percentage points. InceptionV3 achieves the best overall performance (Accuracy: 0.38, Macro F1: 0.32) owing to its multi-scale inception modules, which are well-suited to capturing morphological features at varying resolutions in embryo images. VGG-16 and VGG-19 perform comparably to MobileNetV2 despite having significantly more parameters, suggesting that the bottleneck is the visual similarity of stages rather than model capacity. MobileNetV2, being the most lightweight model, offers the best accuracy-to-parameter trade-off.

The detailed per-class performance of MobileNetV2 under the custom loss is shown in Table 3. The model achieves higher precision on dominant classes such as *tPNa* (0.63) and *t9+* (0.55), but struggles on low-sample or visually similar stages such as *t3* (F1: 0.08) and *t6* (F1: 0.06), reflecting the inherent class imbalance and visual ambiguity of intermediate developmental stages. Importantly, under the custom loss, the most severe

misclassifications — where a stage is predicted as a temporally distant stage — are reduced compared to the baseline, which is the direct effect of the ordinal penalty term $\mathcal{L}_{\text{ordinal}}$.

Table 3: Per-class precision, recall, and F1-score for MobileNetV2 on the test set.

Stage	**Precision**	**Recall**	**F1-score**
tPB2	0.20	0.45	0.28
tPNa	0.63	0.46	0.53
tPNf	0.14	0.38	0.21
t2	0.37	0.36	0.36
t3	0.07	0.09	0.08
t4	0.34	0.33	0.34
t5	0.14	0.13	0.13
t6	0.07	0.06	0.06
t7	0.16	0.07	0.10
t8	0.29	0.39	0.33
t9+	0.55	0.35	0.43
tM	0.27	0.37	0.31
tSB	0.28	0.29	0.28
tB	0.32	0.30	0.31
tEB	0.29	0.29	0.29
Accuracy		0.33	
Macro Avg	0.27	0.29	0.27
Weighted Avg	0.36	0.33	0.33

Discussion

The moderate accuracy range of 33–38% on a 15-class fine-grained classification task is consistent with the inherent difficulty of the problem. Embryo developmental stages are visually subtle and temporally continuous, making boundary stages hard to distinguish even for trained embryologists. The consistent improvement of 3–4% in both accuracy and Macro F1 when using the custom loss over the SCCE baseline across all four architectures demonstrates the benefit of encoding domain knowledge directly into the loss function.

The confidence penalty term $\mathcal{L}_{\text{confidence}}$ forces the model to assign high-probability mass to the correct class rather than spreading uncertainty across many classes, producing sharper and more reliable predictions. The ordinal penalty term $\mathcal{L}_{\text{ordinal}}$ is biologically meaningful — confusing *t2* for *tHB* is far more clinically problematic than confusing *t2* for *t3*. By penalizing distant misclassifications more heavily, the custom loss improves the clinical safety of predictions even when overall accuracy is moderate.

InceptionV3 benefits most from the custom loss due to its multi-scale feature extraction, which aligns well with the varying sizes of embryo cellular structures across stages. VGG models, despite their large parameter counts, show diminishing returns compared to the lighter MobileNetV2, suggesting the visual similarity of stages is the true bottleneck. Future improvements could include temporal sequence modeling using LSTMs or 3D convolutions over frame sequences, self-supervised pre-training on unlabeled embryo data, and focal loss integration to further address class imbalance in rare stages such as *t3* and *tHB*.

Conclusion

This work presents a transfer learning pipeline for embryo stage classification using MobileNetV2, InceptionV3, VGG-16, and VGG-19, combined with a custom total loss function defined as the sum of Sparse Categorical Cross-Entropy, a confidence penalty, and an ordinal stage-distance penalty. The custom loss satisfies all desirable properties — non-negativity, zero at optimum, monotonicity, differentiability, and domain awareness — making it theoretically well-founded and clinically meaningful for the ordered multi-class embryo staging problem.